

INSTALLATION	
<code>curl -LsSf https://astral.sh/</code>	Linux / macOS (recommended)
<code>powershell -c irm astral.sh/u</code>	Windows PowerShell
<code>pip install uv</code>	Via pip — any platform
<code>brew install uv</code>	Homebrew (macOS)
PYTHON MANAGEMENT	
<code>uv python install 3.12</code>	Install Python 3.12
<code>uv python install 3.11 3.12</code>	Install multiple versions
<code>uv python list</code>	List available Pythons
<code>uv python pin 3.12</code>	Pin to .python-version
<code>uv python find</code>	Show active Python path
<code>uv python uninstall 3.11</code>	Remove a version
VIRTUAL ENVIRONMENTS	
<code>uv venv</code>	Create .venv in current dir
<code>uv venv --python 3.12</code>	With specific Python
<code>uv venv myenv</code>	Custom env name
<code>source .venv/bin/activate</code>	Activate (Linux/macOS)
<code>.venv\Scripts\activate</code>	Activate (Windows)
<code>deactivate</code>	Deactivate environment
PIP INTERFACE (UV PIP ...)	
<code>uv pip install requests</code>	Install a package
<code>uv pip install 'django>=4.2'</code>	With version constraint
<code>uv pip install -r requirement</code>	From requirements file
<code>uv pip install -e .</code>	Editable / dev install
<code>uv pip uninstall requests</code>	Remove a package
<code>uv pip list</code>	List installed packages
<code>uv pip freeze</code>	Freeze format output
<code>uv pip show requests</code>	Package metadata
<code>uv pip compile reqs.in -o req</code>	Resolve → requirements
<code>uv pip sync requirements.txt</code>	Exact-sync environment
RUNNING SCRIPTS	
<code>uv run script.py</code>	Run in project env
<code>uv run --with requests script</code>	Inject dep, no install
<code>uv run --python 3.12 script.p</code>	Override Python version
<code>uv run pytest</code>	Run tool from env
<code>uv run python -m http.server</code>	Module execution
KEY GLOBAL FLAGS	
<code>--no-cache</code>	Skip cache entirely
<code>--offline</code>	No network access
<code>--frozen</code>	Don't update lock file
<code>--system</code>	Use system Python
<code>--python VER</code>	Override Python version
<code>--index URL</code>	Custom package index
<code>--quiet / -v</code>	Less / more output

PROJECT MANAGEMENT (UV INIT / ADD / SYNC)	
<code>uv init myproject</code>	Scaffold new project
<code>uv init --lib mylib</code>	Library project
<code>uv init --app myapp</code>	Application project
<code>uv add requests</code>	Add dependency
<code>uv add 'httpx>=0.27'</code>	Add with constraint
<code>uv add --dev pytest ruff</code>	Add dev dependencies
<code>uv add --optional extra pkg</code>	Optional dep group
<code>uv remove requests</code>	Remove dependency
<code>uv sync</code>	Install from lock file
<code>uv sync --frozen</code>	Strict sync, no update
<code>uv sync --no-dev</code>	Skip dev dependencies
<code>uv lock</code>	Update lock without sync
<code>uv lock --upgrade-package htt</code>	Upgrade one package
<code>uv lock --upgrade</code>	Upgrade all packages
<code>uv tree</code>	Show dependency tree
<code>uv export -o requirements.txt</code>	Export to requirements
TOOL RUNNING (UVX / UV TOOL)	
<code>uvx ruff check .</code>	Run ruff ephemerally
<code>uvx black src/</code>	Run black without install
<code>uvx --from 'ruff==0.4' ruff</code>	Pin specific version
<code>uvx --python 3.12 tool</code>	With Python override
<code>uv tool install ruff</code>	Install tool globally
<code>uv tool list</code>	List installed tools
<code>uv tool upgrade ruff</code>	Upgrade a tool
<code>uv tool uninstall ruff</code>	Remove a tool
<code>uv tool dir</code>	Show tools directory
BUILD & PUBLISH	
<code>uv build</code>	Build sdist + wheel
<code>uv build --wheel</code>	Wheel only
<code>uv build --sdist</code>	Source dist only
<code>uv publish</code>	Publish to PyPI
<code>uv publish --index testpypi</code>	Publish to TestPyPI
<code>uv publish --token \$TOKEN</code>	With API token
WORKSPACE (MONOREPO)	
<code>uv init --workspace</code>	Enable workspace in root
<code>uv add --workspace member</code>	Add workspace member
<code>uv sync --package mylib</code>	Sync specific member
<code>uv run --package mylib pytest</code>	Run in specific member

ENVIRONMENT VARIABLES								
UV_PYTHON	UV_INDEX_URL	UV_EXTRA_INDEX_	UV_CACHE_DIR	UV_NO_CACHE=1	UV_SYSTEM_PYTHO	UV_VIRTUAL_ENV	UV_TOOL_DIR	UV_PROJECT_ENVI
Default Python	Primary index	Extra indexes	Cache path	Disable cache	Use system Python	Override venv path	Tools directory	Project venv path